

Project Smokey

Edge-to-cloud wildfire smoke detection on live California camera infrastructure

Daksh Kumar · Built Aug 29 – Sep 17, 2026 · Solo project · Technical brief as of Sep 5, 2026

Summary

A two-stage wildfire smoke detection system. A quantized YOLOv11 detector runs on a Raspberry Pi 5 at the edge, polling live public camera feeds from the HPWREN network across Southern California. A temporal reasoning layer converts noisy per-frame detections into alarm decisions, which are posted to a FastAPI + PostgreSQL + S3 backend.

The core engineering problem is not detection accuracy in isolation. It is producing a usable alarm from a deliberately small model, running on a \$60 device, against a live camera network, with a false-alarm rate low enough that a human would keep the system switched on.

System Architecture — Two-Stage Design

The system is explicitly structured as fast reflexive perception feeding slow deliberate judgment. This division is the central design decision and it is what makes a weak per-frame detector viable in production.

System 1 — per-frame perception (fast, reflexive)

- YOLOv11n object detector, single class, running once per frame
- Answers only: *is there something smoke-shaped here, and how confident am I?*
- Deliberately permissive. Tuned to a low confidence floor and allowed to be wrong often.
- Stateless. Holds no memory between frames.
- Measured at 4.5 ms/frame on a Tesla T4; target budget on the Pi is 60 s/frame

System 2 — temporal reasoning (slow, deliberate)

- Sliding-window aggregator implementing **K-of-M**: an alarm fires only when at least K of the last M frames exceed the confidence threshold
- Exploits the key physical asymmetry: **real smoke persists and grows across frames; spurious detections do not**. A single-frame false positive cannot raise an alarm.
- Carries an `armed` state so one sustained false alarm is counted once rather than once per frame
- Turns per-frame recall of 0.643 into a materially higher per-fire detection rate, because a single fire presents dozens of independent opportunities across a sequence
- Search space of (threshold, K, M) is swept exhaustively; the operating point is selected by minimising median time-to-detection subject to a hard false-alarm budget

Why this matters: raw detector metrics (mAP, per-frame precision/recall) do not describe system behaviour. The quantity that matters operationally is *false alarms per day* versus *fraction of fires detected* versus *minutes from ignition to alarm*. System 2 is where those are actually produced, and the evaluation harness is built around them rather than around mAP.

Data

Source	Role	Resolution	Detail
pyro-sdis (PyroNear)	Training	1280×720 16:9, 0.92 MP	23,695 train images (3,977 backgrounds) / 5,842 val (768 backgrounds), 5,395 val instances. Real bounding boxes. 100%

			French imagery (sdis-07, force-06, sdis-77). Sourced via HuggingFace <code>datasets</code> .
FlgLib (HPWREN/UCSD)	Benchmark	2048×1536 4:3, 3.1 MP	522 California fire sequences; 25 downloaded. Ships zero bounding boxes . Frame-level ground truth encoded in filenames as <code><unix_ts>_<offset>.jpg</code> , offset in seconds relative to ignition (negative = pre-ignition, guaranteed smoke-free).
HPWREN live	Deployment	3072×2048 3:2, 6.3 MP	503 cameras across 81 sites, 199 colour. Live endpoint <code>cdn.hpwren.ucsd.edu/RT/<cam>.jpg</code> . Publish cadence verified at exactly 1 frame/min (180 frames per 3-hour block, deltas 59–61 s).

The two constraints that shape the whole project

Domain gap. The only data with bounding boxes is French. The deployment target is California. The model is trained on one continent and evaluated on another; FlgLib is the honest test precisely because its frames come from the same physical cameras the live system polls.

Resolution gap. Live frames carry 6.8× the pixel count of the training data at a third aspect ratio. At `imgsz=1024` a 3072×2048 frame is compressed roughly 3×, so a 30 px plume reaches the model as ~10 px. 38% of training boxes already occupy under 0.1% of frame area. This is the measured argument for tiling.

Model & Training

Property	Value
Architecture	YOLOv11n, 181 layers, 2,590,035 parameters, 6.5 GFLOPs
Detection head	Single class, three feature scales (64/128/256 channels)
Input size	1024 × 1024
Batch / optimiser	16 / MuSGD (auto-selected), lr 0.01, momentum 0.9
Augmentation	Mosaic (closed at epoch 20), Albumentations: Blur, MedianBlur, ToGray, CLAHE
Hardware	Tesla T4, 15 GB — Kaggle. 5.86 GB peak GPU memory
Run	30 epochs in 4.69 h; 1,481 iterations/epoch at ~2.9 it/s

Results (pyro-sdis validation, French domain)

Precision	Recall	mAP@50	mAP@50-95	Inference
0.727	0.643	0.711	0.441	4.5 ms (T4)

Findings. Best checkpoint was epoch 17; mAP@50 peaked at 0.717 by epoch 15 and oscillated flat through epoch 30, so roughly half the run bought nothing — subsequent runs use 18 epochs. Closing mosaic augmentation produced no measurable bump, which alongside the early plateau indicates the nano backbone is capacity-limited rather than under-trained. The mAP@50 to mAP@50-95 gap (0.711 vs 0.441) reflects loose localisation, expected given smoke has genuinely diffuse boundaries and largely irrelevant to an alerting task.

Note on interpretation. These are French-domain numbers and are treated as a ceiling, not a prediction. The operative metric is the FlgLib cross-continent result.

Evaluation Harnesses

Two independent evaluators, because the two datasets carry fundamentally different label types. This separation is deliberate and is the reason the project can be honestly assessed at all.

	harness.py	figlib_eval.py
Data	pyro-sdis validation	FlgLib sequences
Labels	Bounding boxes	Frame-level, parsed from filenames
Question	Is the box in the right place?	Did the alarm fire, and when?
Implements	IoU; YOLO-normalised to pixel conversion; greedy confidence-ordered matching with claim tracking (duplicates become FP, unclaimed truths become FN); 99-point threshold sweep; operating point at fixed FP budget; box-area size bucketing	Filename offset parsing; sequence ordering; per-frame max-confidence table with CSV caching; K-of-M firing logic with armed-state tracking; false alarms per day; per-fire detection rate; median time-to-detection; 3-dimensional (threshold $\times$ K $\times$ M) sweep; operating point minimising TTD subject to FP budget and minimum detection rate

Edge Deployment

Hardware

- Raspberry Pi 5, 4 GB — 2.4 GHz quad-core Arm Cortex-A76
- Official Active Cooler (sustained inference thermally throttles a passively cooled Pi 5)
- 27 W USB-C PD supply, 64 GB SanDisk microSD
- Raspberry Pi OS Lite 64-bit, headless via SSH — 64-bit required for aarch64 ONNX Runtime

Device program

- Fetch $\rightarrow$ infer $\rightarrow$ queue $\rightarrow$ post
- Polls `/RT/<camera>.jpg` at 1 frame/min/camera, matched exactly to HPWREN's own publish rate. HPWREN is NSF-funded research infrastructure whose terms request non-wasteful bandwidth use; the poll rate is a deliberate compliance decision, and backfills remain sequential.
- Camera liveness determined by `Last-Modified` freshness on the live endpoint rather than the manifest's `active` flag, which marks all 503 cameras live and is therefore useless as a signal
- Local SQLite queue with retry, so detections survive network loss and the device degrades gracefully rather than dropping alerts
- ONNX export with post-training quantisation for CPU inference

Cloud service

- FastAPI ingestion endpoint, PostgreSQL for detection records, S3 for frame storage
- Integrated into a personal orchestration platform (Alfred) at `/api/smokey`

Technology Stack

Languages: Python 3.12 SQL Bash

ML / CV: PyTorch 2.10 Ultralytics YOLOv11 ONNX Runtime CUDA 12.8 Albumentations HuggingFace datasets
NumPy Pillow OpenCV matplotlib

Backend / infra: FastAPI PostgreSQL SQLite Amazon S3 Docker REST APIs

Edge / systems: Raspberry Pi 5 Linux (Pi OS Lite 64-bit) SSH / headless ops ARM64 / aarch64 Model quantisation

Tooling: Kaggle (Tesla T4) Jupyter Git Executable test harnesses CSV result caching

Status

Component	Status	Notes
Data acquisition & audit	Complete	Label format verified, splits confirmed, three-resolution problem measured
HPWREN access mapping	Complete	Live/archive/index routes verified; 3-hour index lag documented; cadence confirmed
Box-level eval harness	Complete	IoU, matching, sweep, operating point, size buckets
FlgLib timeline scorer	Complete	186-line test suite covering offset parsing, K-of-M firing, scoring
Detector training	In progress	Full 30-epoch run completed and characterised; re-running at 18 epochs
Cross-continent benchmark	Next	FlgLib runner built; produces the project's headline result
Tiling	Next	Overlapping grid; full-frame vs tiled A/B on identical sequences
ONNX export + quantisation	Planned	
Edge device program	Planned	Fetch loop, SQLite queue, retry, POST
Cloud service	Planned	FastAPI + Postgres schema + S3
Thermal benchmarking	Planned	Sustained inference under load on the Pi

Planned Technical Work

Tiling

Overlapping tile grid over full-resolution frames so small plumes reach the model at native scale instead of being destroyed by downscaling. Guarantees full pixel coverage, uniform tile size, and edge-shifted rather than truncated boundary tiles. Verified by an executable spec before any number depending on it is trusted. For frame-level FlgLib scoring, per-tile confidences collapse to a max, so no box merging is required; coordinate remapping and seam de-duplication are needed only for the live system, which must report location.

Adaptive thresholding

Per-camera, per-hour dynamic threshold derived from the same camera's average prediction at the same time of day over prior days, rather than one global cutoff. Directly targets periodic confounders — fog, marine layer, solar reflection, haze — that recur at fixed times on fixed cameras. Approach adapted from UCSD's SmokeyNet.

Free in-domain negatives

FlgLib frames with negative offsets are guaranteed smoke-free California imagery, labelled by filename and requiring zero annotation. Adding them as training backgrounds suppresses false positives in the deployment domain, which buys threshold headroom, which converts directly into recall at a fixed alarm budget. This is the only mechanism available for getting target-domain data into training, since FlgLib carries no boxes.

Capacity scaling

At 1 frame/min the inference budget is 60 seconds per frame against a nano model measured at 4.5 ms on a T4, leaving very large headroom even after the CPU penalty on ARM. Larger backbones (YOLOv11s/m) are therefore viable and are indicated by the observed capacity ceiling. Tiling and model size compete for the same budget; the trade-off is to be measured rather than assumed.

Prior Art & Positioning

- **ALERTCalifornia / DigitalPath** — the production system for California. 1,200+ PTZ cameras, 8–16 M images/day, deployed with CAL FIRE since 2023; informed ~3,600 incidents in 2025. Proprietary and closed.
- **SmokeyNet (UCSD)** — published spatio-temporal baseline on FlgLib, field-tested across 65 HPWREN cameras. Source of the adaptive-threshold approach.
- **PyroNear** — open-source edge wildfire detection on Raspberry Pi with ONNX inference and an alert platform; author of the pyro-sdis training set. Closest architectural analogue, and a reference implementation for the edge loop.

Engineering Principles Applied

- **Measure before optimising.** No tuning against the deployment domain until the cross-continent baseline exists; otherwise the fix is a guess.
- **Metrics must match the decision.** mAP does not describe alarm behaviour; the harness reports false alarms/day, detection rate, and time-to-detection instead.
- **Test geometry and scoring logic without a model.** Both evaluators and the tiling grid are verified against synthetic inputs, so correctness bugs cannot hide inside model error.
- **Honest evaluation over flattering evaluation.** The benchmark was chosen to be hard and out-of-domain, and the expected performance drop is the finding, not a failure.
- **Respect the shared resource.** Poll rate matched to the upstream publish rate by design on NSF-funded public infrastructure.

Project Smokey · Daksh Kumar · Technical brief generated Sep 5, 2026. Numbers reflect the completed 30-epoch pyro-sdis training run; cross-continent FlgLib results pending.